

Supplementary Summer Examination – 2023

Course: B. Tech.

Branch :Civil Engineering

Semester :VII

Subject Code & Name:BTCVC701

Design of Concrete Structures-II

Max Marks: 60

Date:13/07/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Use of IS 456 and IS 1343 is allowed

(Level/CO) Marks

Q.1 Solve Any Two of the following.

12

- | | | | |
|----|---|-----|---|
| A) | Explain in detail- Equilibrium torsion and Compatibility torsion | CO1 | 6 |
| B) | A RC beam of rectangular section 300mm wide and 600mm overall deep is reinforced with 4 bars of 25mm diameter distributed at each of the corners at an effective cover of 50mm in the direction of depth and side cover of 25mm in the direction of width. 8 mm diameter two-legged stirrups are provided at spacing of 100mm centre to centre. Estimate the torsional strength of the section adopting Fe415 HYSD bars and M20 concrete for following cases.
i) Torsional strength if $V_u = 0$
ii) Torsional strength if $V_u = 100\text{kN}$. | CO1 | 6 |
| C) | Design a axially loaded short column 400*400mm with minimum eccentricity of 20mm for a service load of 1500kN. Use M20 and Fe415 | CO2 | 6 |

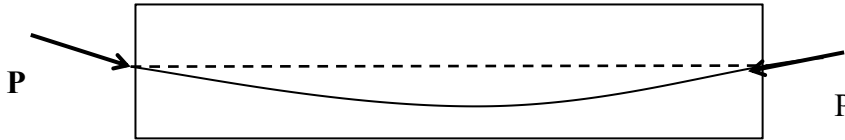
Q.2 Solve Any Two of the following.

12

- | | | | |
|----|--|-----|---|
| A) | Beam of size 400*500 mm has to resist a factored torsional moment of 18kNm and a factored bending moment of 100kNm in addition to shear force of 180kN. Design the longitudinal reinforcement ONLY. Use M20 and Fe415, 25mm dia main longitudinal bars with 25mm clear cover, 12 mm dia stirrups | CO1 | 6 |
| B) | Design the reinforcement in circular column of 400mm dia subjected to a factored load of 1800kN. The column has an unsupported length of 3.4m. use M25 and Fe415 steel. Assume clear cover of 40mm over spirals of 6mm dia | CO2 | 6 |
| C) | Enlist the Functions of longitudinal and transverse reinforcement in column. | CO2 | 6 |

Q. 3 Solve Any One of the following.**12**

- A) Calculate stresses at QUARTER span in prestressed concrete beam as shown in figure by using three methods of prestress analysis (stress concept, load balancing concept and pressure line method) at mid span (without loss ratio). The beam is subjected to a live load of 70kN/m. $P=1600\text{kN}$, Length of beam=8m, Eccentricity of cable at centre=200mm. Beam size=400mm wide and 600mm deep. unit weight of concrete is 25kN/m^3

CO3**12**

- B) An unsymmetrical prestressed concrete section has top flange 500mm*200mm, bottom flange 400mm*300mm and web 200mm*800mm, it is supported over a span of 15m carries live load of 13kN/m, the effective prestressing force is 1000kN located at 100mm from soffit of the section at mid span, cable profile is parabolic and concentric at support. Calculate extreme fiber stresses in concrete at mid span at final stage **ONLY**, by using stress concept method. Loss ratio is 0.85, unit weight of concrete is 25kN/m^3

CO3**12****Q.4 Solve Any Two of the following.****12**

- A) Write a short note on Structural Audit
- B) Explain in detail- Losses in prestress
- C) A pre-tensioned concrete beam has a cross section 250mm*360mm prestressed by a cable having a cross sectional area of 502 mm^2 . The centroid of the cable is located at 105mm from the soffit. Find the percentage loss of stress in steel due to creep, shrinkage, elastic deformation of concrete and 5% relaxation of stress in steel. The initial prestress in steel is 1000 Mpa. $E_s=210\text{Gpa}$, creep coeff=1.6, $E_c=36.8\text{Gpa}$, shrinkage strain= 1.9×10^{-4}

CO4**6****CO3****6****CO3****6****Q. 5 Solve the following questions.****12**

- A) Explain in brief why high grade materials are used for pre-stressed concrete elements?
- B) The end block of a post tensioned prestressed concrete bridge girder is of rectangular section 500mm wide and 1500mm deep freyssinet anchorage of 7-15mm dia type anchor of size 250mm*250mm are used. Three anchorage are provided with space of 500mm centre to centre. The jacking force in each anchorage is 1800kN. Design a suitable anchorage zone reinforcement in the end block. $F_y=415\text{N/mm}^2$

CO3**6****CO4****6******* End *****